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Switching control circuit and power supply circuit

Abstract

A switching control circuit for a power supply circuit that includes a transformer including primary and secondary coils, and first and second transistors controlling a current flowing through the primary coil. The power supply circuit generates an output voltage at a target level. The switching control circuit controls switching of the first and second transistors. The switching control circuit includes: a drive signal output circuit configured to output drive signals according to a normal mode, when the output voltage is lower than a first level, and output the drive signals according to a burst mode, when the output voltage is higher than the first level; and a driver circuit configured to switch the first and second transistors in response to the drive signals. The drive signal output circuit reduces a first time period in the burst mode, in response to the output voltage rising above the first level.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority pursuant to 35 U.S.C. § 119 from Japanese patent

BACKGROUND

Technical Field

(2) The present disclosure relates to a switching control circuit and a power supply circuit.

Description of the Related Art

(3) Power supply circuits generating an output voltage at a target level from an input voltage have been disclosed (for example, Japanese Patent No. 6787505, WO2019/155733, Japanese Patent Application Publication Nos. 2013-26079, 2014-108004, 2017-112798, 2021-93813, and 2021-93814, and U.S. patent Ser. No. 10/069,403).

(4) When a load of a power supply circuit enters a light condition, a control circuit of the power supply circuit may switch transistors in a burst mode from a normal mode. In this event, the operation of the power supply circuit may become unstable when a method of switching the transistors in the normal mode is different from a method of switching the transistors in the burst mode.

SUMMARY

(5) A first aspect of an embodiment of the present disclosure is a switching control circuit for a power supply circuit that includes a transformer including a primary coil and a secondary coil, and first and second transistors configured to control a current flowing through the primary coil, the power supply circuit being configured to generate an output voltage at a target level, the switching control circuit configured to control switching of the first and second transistors, the switching control circuit comprising: a drive signal output circuit configured to output drive signals according to a normal mode, when the output voltage is lower than a first level, and output the drive signals according to a first burst mode, when the output voltage is higher than the first level; and a driver circuit configured to switch the first and second transistors in response to the drive signals, wherein the normal mode is a mode in which the first and second transistors are continuously switched, in a first cycle based on a feedback voltage corresponding to the output voltage, the first burst mode is a mode in which a first time period, during which the first and second transistors are continuously switched in the first cycle, and a second time period, during which the switching of the first and second transistors is stopped, are repeated in a predetermined cycle, and the drive signal output circuit reduces the first time period in the predetermined cycle, in response to the output voltage rising above the first level.

(6) A second aspect of an embodiment of the present disclosure is a power supply circuit configured to generate an output voltage at a target level, the power supply circuit comprising: a transformer including a primary coil and a secondary coil; first and second transistors configured to control a current flowing through the primary coil; and a switching control circuit configured to control switching of the first and second transistors, wherein the switching control circuit includes: a drive signal output circuit configured to output drive signals according to a normal mode, when the output voltage is lower than a first level, and output the drive signals according to a first burst mode, when the output voltage is higher than the first level, and a driver circuit configured to switch the first and second transistors, in response to the drive signals, wherein the normal mode is a mode in which the first and second transistors are continuously switched, in a first cycle based on a feedback voltage corresponding to the output voltage, and the first burst mode is a mode in which a first time period, during which the first and second transistors are continuously switched in the first cycle, and a second time period, during which the switching of the first and second transistors is stopped, are repeated in a predetermined cycle; and the drive signal output circuit reduces the first time period in the predetermined cycle, in response to the output voltage rising above the first level.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram illustrating an example of a DC-DC converter **10**.
- (2) FIG. 2 is a diagram illustrating an example of a control IC **50**.
- (3) FIG. 3 is a diagram illustrating an example of a detection circuit **140**.
- (4) FIG. 4 is a chart illustrating an example of voltages V_{wd} , V_{c1} , and V_{c2} .
- (5) FIG. 5 is a chart for explaining a relationship between operation modes of a digital control circuit **150** and various voltages.
- (6) FIG. 6 is a chart for explaining operation modes of a digital control circuit **150**.
- (7) FIG. 7 is a chart illustrating an example of drive signals V_{dr1} and V_{dr2} in a normal mode **M1**.
- (8) FIG. 8 is a chart for explaining a switching cycle T_{sw} of drive signals V_{dr1} and V_{dr2} .
- (9) FIG. 9 is a chart for explaining a relationship between a gain of a DC-DC converter **10** and a switching frequency.
- (10) FIG. 10 is a chart for explaining a switching period $T1$ and a stop period $T2$ in a burst mode **M2**.
- (11) FIGS. 11A to 11C are charts for explaining changes in a switching period $T1$ in a burst mode **M2**.
- (12) FIG. 12 is a chart for explaining a switching period $T1$ and a stop period $T2$ in a burst mode **M3**.
- (13) FIG. 13 is a chart illustrating an example of changes in operation mode of a DC-DC converter **10**.

DETAILED DESCRIPTION

- (14) At least following matters will become apparent from the descriptions of the present specification and the accompanying drawings.
 - (15) Herein, the same or equivalent constituent elements, members and the like illustrated in the drawings are given the same reference signs, and redundant description thereof is omitted for convenience. In an embodiment of the present disclosure, the term “couple” means to electrically couple unless otherwise noted. Thus, “coupling” includes a case where two components are coupled not only through wiring but also through a resistor, for example.
- Embodiment
- (16) <<<Overview of DC-DC Converter **10**>>>
 - (17) FIG. 1 is a diagram illustrating an example of a DC-DC converter **10**. The DC-DC converter **10** is an LLC current resonant power supply circuit to generate an output voltage V_{out} at a target level (e.g., 12 V) from a predetermined input voltage V_{in} (e.g., 400 V).
 - (18) The DC-DC converter **10** also applies the output voltage V_{out} to a load **11**. Here, it is assumed that the load **11** is an LED of a lighting device (not illustrated), for example, and a current flowing through the load **11** is a load current I_{out} .
 - (19) The DC-DC converter **10** includes capacitors **30**, **34**, **35**, and **42**, N-channel metal-oxide-semiconductor (NMOS) transistors **31** and **32**, a transformer **33**, a resistor **36**, a control block **37**, diodes **40** and **41**, a voltage regulator circuit **43**, and a light-emitting diode **44**.
 - (20) The capacitor **30** stabilizes a voltage between a power supply line to receive the input voltage V_{in} and a ground line on the ground side, to remove noise and the like.
 - (21) The NMOS transistor **31** is a high-side power transistor, and the NMOS transistor **32** is a low-side power transistor. The NMOS transistors **31** and **32** are used as switching devices in an embodiment of the present disclosure, however, for example, P-channel metal-oxide-semiconductor (PMOS) transistors or bipolar transistors may be used.
 - (22) The transformer **33** includes a primary coil **L1**, secondary coils **L2** and **L3**, and an auxiliary coil **L4**. The primary coil **L1**, the secondary coils **L2** and **L3**, and the auxiliary coil **L4** are insulated

from one another. In the transformer **33**, voltages are generated at the secondary coils **L2** and **L3** on the secondary side according to changes in voltage across the primary coil **L1** on the primary side, and a voltage is generated at the auxiliary coil **L4** on the primary side according to changes in voltage of the secondary coils **L2** and **L3**.

(23) The primary coil **L1** has one end coupled to a source of the NMOS transistor **31** and a drain of the NMOS transistor **32**, and the other end coupled to a source of the NMOS transistor **32** through the capacitor **34**. Accordingly, in response to switching of the NMOS transistors **31** and **32** being started, the voltages at the secondary coils **L2** and **L3** and the auxiliary coil **L4** change.

(24) In an embodiment of the present disclosure, the primary coil **L1** and the secondary coils **L2** and **L3** are electromagnetically coupled with the opposite polarities, and the secondary coils **L2** and **L3** and the auxiliary coil **L4** are electromagnetically coupled with the opposite polarities. The polarities of the electromagnetic coupling of the primary coil **L1**, the secondary coils **L2** and **L3**, and the auxiliary coil **L4** of the transformer **33** are not limited to the above.

(25) The capacitor **34** is a so-called resonant capacitor configuring a resonant circuit, with the primary coil **L1** of the transformer **33** and a leakage inductance. Note that, in FIG. **1**, illustration of the leakage inductance is omitted.

(26) The capacitor **35** and the resistor **36** coupled in series with the capacitor **35** are circuits coupled in parallel with the capacitor **34** to shunt a resonant current of the resonant circuit. In an embodiment of the present disclosure, a current I_s obtained by shunting the resonant current flows through the capacitor **35** and the resistor **36**, thereby generating a voltage V_s corresponding to the resonant current at a coupling node between the capacitor **35** and the resistor **36**.

(27) It is assumed in an embodiment of the present disclosure that a direction of a current flowing from the primary coil **L1** to the capacitor **34** is a “positive direction”, and a direction of a current flowing from the capacitor **34** to the primary coil **L1** is a “negative direction”.

(28) The control block **37** is a circuit block to control switching of the NMOS transistors **31** and **32**, and will be described later in detail.

(29) The diodes **40** and **41** rectify the voltages at the secondary coils **L2** and **L3**, and the capacitor **42** smooths the rectified voltage. As a result, the smoothed output voltage V_{out} is generated at the capacitor **42**. Note that the output voltage V_{out} reaches a direct-current voltage (DC) voltage at a target level (e.g., 12 V).

(30) The voltage regulator circuit **43** generates a constant DC voltage, and is configured using a shunt regulator, for example.

(31) The light-emitting diode **44** is a device to emit light having an intensity corresponding to a difference between the output voltage V_{out} and an output of the voltage regulator circuit **43**, to configure a photocoupler together with a phototransistor **65**, which will be described later. In an embodiment of the present disclosure, the intensity of the light from the light-emitting diode **44** increases as the level of the output voltage V_{out} rises.

(32) ===Control Block **37**===

(33) The control block **37** includes a control IC **50**, a diode **60**, capacitors **61**, **62**, **66**, and **67**, resistors **63** and **64**, and a phototransistor **65**.

(34) The control IC **50** is an integrated circuit to control switching of the NMOS transistors **31** and **32**, and has terminals VCC, BO, FB, VW, FB, IS, CA, HO, and LO. The control IC **50** has a ground terminal to receive a ground voltage, and the like, for example, which are omitted here for convenience.

(35) The terminal VCC is a terminal to receive a voltage V_{cc} to operate the control IC **50**. A cathode of the diode **60** and the capacitor **61** having one end that is grounded are coupled to the terminal VCC. The auxiliary coil **L4** is coupled to an anode of the diode **60**. In an embodiment of the present disclosure, when the control IC **50** starts switching the NMOS transistors **31** and **32**, a voltage corresponding to the load current I_{out} is generated at the auxiliary coil **L4**.

(36) As a result, the capacitor **61** is charged with the current from the diode **60**, and the charge

voltage of the capacitor **61** results in a power supply voltage V_{cc} to operate the control IC **50**. Note that the control IC **50** is started up based on a predetermined voltage (e.g., alternating current (AC) voltage from a commercial power supply) applied to a terminal not illustrated, and then operates based on the power supply voltage V_{cc} .

(37) The terminal VW is a terminal to detect the voltage at the auxiliary coil L4. In an embodiment of the present disclosure, the capacitor **62** and resistors **63** and **64** that are coupled in series are coupled in parallel with the auxiliary coil L4. The capacitor **62** is an element allowing an AC component of the voltage from the auxiliary coil L4 to pass therethrough.

(38) The resistors **63** and **64** configure a voltage divider circuit to divide the AC component of the voltage generated at the auxiliary coil L4. In an embodiment of the present disclosure, a node at which the resistors **63** and **64** are coupled is coupled to the terminal VW, and the voltage at the terminal VW results in a voltage V_w .

(39) The terminal BO is a terminal to receive the input voltage V_{in} of the DC-DC converter **10**.

(40) The terminal FB is a terminal at which a feedback voltage V_{fb} corresponding to the output voltage V_{out} is generated, and to which the phototransistor **65** and the capacitor **66** are coupled. The phototransistor **65** passes a bias current I_1 corresponding to the intensity of the light from the light-emitting diode **44**, from the terminal FB to the ground. The capacitor **66** is provided to remove noise between the terminal FB and the ground. Accordingly, the phototransistor **65** operates as a transistor to generate a sink current.

(41) The terminal IS is a terminal to receive a voltage corresponding to the resonant current of the DC-DC converter **10**. As described above, the voltage V_s corresponding to the current I_s obtained by shunting the resonant current is generated at the node at which the capacitor **35** and the resistor **36** are coupled. In an embodiment of the present disclosure, the node at which the capacitor **35** and the resistor **36** are coupled is coupled to the terminal IS, and thus the voltage V_s is applied to the terminal IS.

(42) The input voltage V_{in} is received through the terminal BO and detected inside the control IC **50**. This is because power corresponding to the resonant current and the current I_s (e.g., power of the load **11**) is proportional to the input voltage V_{in} even when the resonant current and the current I_s have the same magnitude.

(43) Through the terminal CA, a voltage V_{ca} , which varies with the voltage V_s at the terminal IS and indicates power consumed by the load **11**, is applied to the capacitor **67**.

(44) The terminal HO is a terminal through which a signal V_{o1} to drive the NMOS transistor **31** is outputted, and to which a gate of the NMOS transistor **31** is coupled.

(45) The terminal LO is a terminal through which a signal V_{o2} to drive the NMOS transistor **32** is outputted, and to which a gate of the NMOS transistor **32** is coupled.

(46) The control IC **50** corresponds to a “switching control circuit”, the capacitor **34** corresponds to a “first capacitor”, and the auxiliary coil L4 corresponds to an “auxiliary coil”. The NMOS transistor **31** corresponds to a “first transistor” and the NMOS transistor **32** corresponds to a “second transistor”. The terminal VW corresponds to a “first terminal” and the terminal BO corresponds to a “second terminal”.

(47) <<<Details of Control IC **50**>>>

(48) FIG. 2 is a diagram illustrating a configuration example of the control IC **50**. The control IC **50** includes analog-to-digital (AD) converters (ADCs) **100**, **111**, and **131**, resistors **110**, **120**, and **121**, a comparator **122**, a load detection circuit **130**, a detection circuit **140**, a digital control circuit **150**, and a driver circuit **151**.

(49) ===AD Converter **100**===

(50) The AD converter **100** converts the input voltage V_{in} applied to the terminal BO into a digital value, to thereby output a resultant voltage. In FIG. 2, voltages converted into digital values are also illustrated in the same manner as analog voltages before conversion.

(51) ===Resistor **110** and AD Converter **111**===

(52) The resistor **110** is an element to generate, at the terminal FB, the feedback voltage Vfb corresponding to the bias current I1 in FIG. 1. The resistor **110** has one end to receive a power supply voltage Vdd and the other end coupled to the terminal FB.

(53) In an embodiment of the present disclosure, in response to the output voltage Vout exceeding the target level, the intensity of the light from the light-emitting diode **44** increases and the bias current I1 increases, and thus the feedback voltage Vfb drops. Meanwhile, in response to the output voltage Vout drops below the target level, the intensity of the light from the light-emitting diode **44** decreases and the bias current I1 decreases, and thus the feedback voltage Vfb rises.

(54) The feedback voltage Vfb is expressed by the following equation (1):

$$V_{fb} = V_{dd} - R \times I_1 \quad (1),$$

where R is a resistance value of the resistor **110**.

(55) The AD converter **111** converts the feedback voltage Vfb generated at the terminal FB into a digital value, to thereby output a resultant voltage.

(56) ===Resistors **120** and **121** and Comparator **122**===

(57) The resistors **120** and **121** are so-called level shifter circuits configured to shift the center level of the voltage Vs that changes around the ground voltage (i.e., 0 V). The resistors **120** and **121** of an embodiment of the present disclosure both have the same resistance value. The power supply voltage Vdd is applied to the resistor **120** of the resistors **120** and **121**, which are coupled in series, and the resistor **121** is coupled to the terminal IS.

(58) Accordingly, the resistor **121** is coupled, through the terminal IS, to the coupling node between the capacitor **35** and the resistor **36** in FIG. 1. Here, when the current Is flowing through the resistor **36** is zero, the voltage Vs is zero. Accordingly, in this case, the level of the voltage Vsd at the node at which the resistors **120** and **121** in FIG. 2 are coupled results in Vdd/2.

(59) In response to the current Is flowing in the positive direction, the voltage Vs also results in a positive voltage, and thus the voltage Vsd rises from Vdd/2. Meanwhile, in response to the current Is flowing in the negative direction, the voltage Vs also results in a negative voltage, and thus the voltage Vsd drops from Vdd/2. Accordingly, the voltage Vsd in an embodiment of the present disclosure changes around the level of Vdd/2, according to the direction of the current Is and the magnitude of the current Is.

(60) The comparator **122** is a circuit configured to detect the direction of the current Is, based on the voltage Vsd and a predetermined reference voltage Vref0. In an embodiment of the present disclosure, since the level of the reference voltage Vref0 is Vdd/2, the comparator **122** outputs a voltage Vc0 at a high level (hereinafter, referred to as high or high level) when the current Is is in the positive direction, and outputs a voltage Vc0 at a low level (hereinafter, referred to as low or low level) when the current Is is in the negative direction.

(61) Accordingly, the resistors **120** and **121** and the comparator **122** of an embodiment of the present disclosure operate as a circuit to detect the direction in which the current Is flows.

(62) ===Load Detection Circuit **130**===

(63) The load detection circuit **130** is a circuit to detect a condition of the load **11** (here, power consumption of the load **11**). To be more specific, the load detection circuit **130** outputs the voltage Vca corresponding to the power of the load **11** to the terminal CA to which the capacitor **67** is coupled, based on the voltage Vsd corresponding to the current Is (i.e., the resonant current).

(64) The AD converter **131** converts the voltage Vca generated at the terminal CA into a digital value, to thereby output a resultant voltage. Note that the voltage Vca increases with an increase in the power consumption of the load **11** and an increase in the current Is, for example. Thus, the digital control circuit **150**, which will be described later, can determine the power consumption of the load **11**, based on the voltage Vca. Note that the power corresponding to the resonant current and the current Is (e.g., the power of the load **11**) is proportional to the input voltage Vin as described above. Thus, a process of correcting the voltage Vca may be performed using the AD-converted input voltage Vin.

(65) ===Detection Circuit **140**===

(66) The detection circuit **140** is a circuit to detect the timing at which the DC-DC converter **10** is operated in a predetermined burst mode (described later), based on the voltage V_w . FIG. **3** is a diagram illustrating an example of the detection circuit **140**. The detection circuit **140** includes resistors **200** and **201**, digital-to-analog (DA) converters (DACs) **210** and **211**, and comparators **212** and **213**.

(67) The resistors **200** and **201** correspond to level shifter circuits configured to shift the center level of the voltage V_w applied to the terminal VW in FIG. **1**. In an embodiment of the present disclosure, it is assumed that a voltage V_{wd} is a voltage generated at the node at which the resistors **200** and **201** are coupled by shifting the level of the voltage V_w .

(68) Here, the voltage at the node at which the auxiliary coil L4 and the capacitor **62** are coupled changes in the positive and negative directions with 0 V as the reference voltage, in response to the NMOS transistors **31** and **32** being switched. As a result, the voltage V_{wd} obtained by level-shifting the voltage V_w has an AC waveform as illustrated in the upper part of FIG. **4**, for example.

(69) The DA converter **210** outputs a reference voltage V_{ref1} at a level corresponding to data D1 from the digital control circuit **150** which will be described later. The DA converter **211** outputs a reference voltage V_{ref2} at a level corresponding to data D2.

(70) The voltage across the auxiliary coil L4 changes according to the level of the input voltage V_{in} . To be more specific, as the level of the input voltage V_{in} rises, a change in the voltage across the auxiliary coil L4 also increases. As a result, an amplitude of the voltage V_{wd} during switching of the NMOS transistors **31** and **32** also increases.

(71) The digital control circuit **150** of an embodiment of the present disclosure adjusts the levels of the reference voltages V_{ref1} and V_{ref2} according to the level of the input voltage V_{in} . Here, in response to the level of the input voltage V_{in} rising, the digital control circuit **150** outputs data D1 and D2 such that the level of the reference voltage V_{ref1} rises and the level of the reference voltage V_{ref2} drops. As a result, the reference voltages V_{ref1} and V_{ref2} are adjusted such that the difference in level between the reference voltages V_{ref1} and V_{ref2} increases.

(72) Meanwhile, in response to the level of the input voltage V_{in} dropping, the digital control circuit **150** outputs data D1 and D2 such that the level of the reference voltage V_{ref1} drops and the level of the reference voltage V_{ref2} rises. As a result, the reference voltages V_{ref1} and V_{ref2} are adjusted such that the difference in level between the reference voltages V_{ref1} and V_{ref2} decreases.

(73) The comparator **212** compares the voltage V_{wd} and the reference voltage V_{ref1} , to thereby output a voltage V_{c1} indicating a result of comparison. To be more specific, the comparator **212** outputs a high voltage V_{c1} , in response to the voltage V_{wd} exceeding the reference voltage V_{ref1} , and outputs a low voltage V_{c1} , in response to the voltage V_{wd} dropping below the reference voltage V_{ref1} .

(74) The comparator **213** compares the voltage V_{wd} and the reference voltage V_{ref2} , to thereby output a voltage V_{c2} indicating a result of comparison. To be more specific, the comparator **213** outputs a low voltage V_{c2} , in response to the voltage V_{wd} exceeding the reference voltage V_{ref2} , and outputs a high voltage V_{c2} , in response to the voltage V_{wd} dropping below the reference voltage V_{ref2} .

(75) Although details will be described later, the digital control circuit **150** outputs high drive signals V_{dr1} and V_{dr2} to turn on the NMOS transistors **31** and **32**, in response to the voltages V_{c1} and V_{c2} going high, respectively, in a predetermined burst mode M3 (described later).

(76) ===Digital Control Circuit **150**===

(77) The digital control circuit **150** in FIG. **2** outputs the drive signals V_{dr1} and V_{dr2} according to the operation mode of the DC-DC converter **10**, based on various voltages inputted thereto. The digital control circuit **150** includes, for example, a digital signal processing circuit (DSP) not illustrated and a storage circuit (e.g., memory).

(78) Although details will be described later, as illustrated in FIG. **5**, the digital control circuit **150**

outputs the drive signals Vdr1 and Vdr2 according to three operation modes, which are a normal mode M1, a burst mode M2, and the burst mode M3, to thereby operate the DC-DC converter 10. (79) Hereinafter, in an embodiment of the present disclosure, a state in which the digital control circuit 150 outputs the drive signals Vdr1 and Vdr2 according to the predetermined operation mode may be referred to as “the digital control circuit 150 operates in the predetermined operation mode”. Moreover, when the digital control circuit 150 operates in the predetermined operation mode, the DC-DC converter 10 also operates in the predetermined operation mode.

(80) Here, the “normal mode” is an operation mode in which the control IC 50 continuously switches the NMOS transistors 31 and 32. The “burst mode” is an operation mode in which the control IC 50 intermittently switches the NMOS transistors 31 and 32. Thus, in the “burst mode”, a switching period (hereinafter referred to as a switching period T1) during which the NMOS transistors 31 and 32 are switched and a stop period (hereinafter referred to as a stop period T2) during which the switching of the NMOS transistors 31 and 32 is stopped are repeated.

(81) As illustrated in FIG. 5, the digital control circuit 150 uses the feedback voltage Vfb and the voltage Vc0 indicating the direction of the current Is, when operating in the normal mode M1 or the burst mode M2. Further, the digital control circuit 150 uses the feedback voltage Vfb, the voltage Vw, and the input voltage Vin, when operating in the burst mode M3. The operation modes of the digital control circuit 150 will be described later in detail.

(82) The digital control circuit 150 configured to output the drive signals Vdr1 and Vdr2 corresponds to a “drive signal output circuit”.

(83) ===Driver Circuit 151===

(84) The driver circuit 151 is a buffer circuit configured to switch the NMOS transistors 31 and 32 in response to the drive signals Vdr1 and Vdr2. To be more specific, the driver circuit 151 drives the NMOS transistor 31 using a signal Vo1 having the same logic level as that of the drive signal Vdr1, and drives the NMOS transistor 32 using a signal Vo2 having the same logic level as that of the drive signal Vdr2.

(85) <<<Operation of Digital Control Circuit 150>>>

(86) FIG. 6 is a chart for explaining the operation of the digital control circuit 150. In FIG. 6, the horizontal axis represents the voltage Vca corresponding to the power consumption of the load 11, and the vertical axis represents the feedback voltage Vfb and the output voltage Vout. As described above, in an embodiment of the present disclosure, when the output voltage Vout rises, the feedback voltage Vfb drops. Accordingly, on the vertical axis of FIG. 6, the direction in which the level of the feedback voltage Vfb rises (upward in FIG. 6) is opposite to the direction in which the output voltage Vout rises (downward in FIG. 6).

(87) In FIG. 6, voltages V0 to V4 on the axis of the output voltage Vout correspond to voltages V10 to V14 on the axis of the feedback voltage Vfb, respectively. That is, in response to the level of the output voltage Vout reaching the level of the voltage V0, the level of the feedback voltage Vfb reaches the level of the voltage V10. Note that the relationships between other voltages V1 to V4 and the voltages V11 to V14 are the same as the relationship between the voltages V0 and V10, and thus detailed description thereof is omitted here. Also, a target level of the output voltage Vout corresponds to the level of a voltage V15 of the feedback voltage Vfb.

(88) First, here, a description is given of a case where the power consumption of the load 11 is greater than 3% when the rated power consumption of the load 11 (hereinafter referred to as “rated power” as appropriate) is 100%. In an embodiment of the present disclosure, a state in which the power consumption of the load 11 is equal to or less than 3% of the rated power is referred to as “the load 11 being in the light load condition”. The digital control circuit 150 determines whether the power consumption of the load 11 is greater than 3%, based on the voltage Vca.

(89) The level of the voltage V1 corresponds to a “first level”, the level of the voltage V2 corresponds to a “second level”, the level of the voltage V3 corresponds to a “third level”, and the level of the voltage V4 corresponds to a “fourth level”. Also, the “first value” corresponds to “3%

of the rated power”, the burst mode M2 corresponds to a “first burst mode”, and the burst mode M3 corresponds to a “second burst mode”.

(90) ===When Load **11** is Heavier Than Light Load Condition===

(91) <<<When Level of Output Voltage V_{out} is Lower Than Voltage V_0 >>>

(92) First, with reference to FIG. 6, a description is given of the digital control circuit **150** when the level of the output voltage V_{out} is lower than the level of the voltage V_0 . The voltage V_0 is a voltage sufficiently lower than the target level (e.g., 12V) of the output voltage V_{out} . For example, when the input voltage V_{in} is lower than a predetermined level, the level of the output voltage V_{out} may become lower than the level of the voltage V_0 .

(93) Such a state in which the level of the output voltage V_{out} is lower than the level of the voltage V_0 may be, hereinafter, referred to as “the output voltage V_{out} being lower than the voltage V_0 ” as appropriate. Further, as described above, in response to the level of the output voltage V_{out} reaching the voltage V_0 , the feedback voltage V_{fb} reaches the voltage V_{10} .

(94) When the output voltage V_{out} is lower than the voltage V_0 and the feedback voltage V_{fb} is higher than the voltage V_{10} , the digital control circuit **150** causes both the drive signals V_{dr1} and V_{dr2} to be low, to stop switching the NMOS transistors **31** and **32**, based on the feedback voltage V_{fb} .

(95) As such, in response to the output voltage V_{out} reaching the voltage V_0 which is sufficiently lower than the target level, the DC-DC converter **10** stops switching the NMOS transistors **31** and **32** so that the DC-DC converter **10** does not become unstable.

(96) Note that, for example, upon startup of the DC-DC converter **10**, the output voltage V_{out} rises from zero. In such a case, the digital control circuit **150** is designed such that the output voltage V_{out} does not stop switching the NMOS transistors **31** and **32**.

(97) <<<When Output Voltage V_{out} is Higher Than Voltage V_0 and Lower Than Voltage V_1 >>>

(98) In response to the output voltage V_{out} exceeding the voltage V_0 , the digital control circuit **150** of FIG. 2 outputs drive signals V_{dr1} and V_{dr2} according to the normal mode M1, based on the feedback voltage V_{fb} , so that the output voltage V_{out} reaches the target level.

(99) FIG. 7 is a chart illustrating an overview of the drive signals V_{dr1} and V_{dr2} according to the normal mode M1. The digital control circuit **150** complementarily changes the drive signals V_{dr1} and V_{dr2} in the normal mode M1. To be more specific, the digital control circuit **150** changes the drive signal V_{dr2} to low, in response to the drive signal V_{dr1} going high, and changes the drive signal V_{dr2} to high, in response to the drive signal V_{dr1} going low.

(100) In FIG. 7, a time period (so-called dead time) during which both the drive signals V_{dr1} and V_{dr2} are low is provided between the drive signals V_{dr1} and V_{dr2} , however, it is omitted here, for convenience.

(101) As illustrated in FIG. 5, the digital control circuit **150** according to an embodiment of the present disclosure determines a switching cycle T_{sw} of the drive signals V_{dr1} and V_{dr2} , based on the feedback voltage V_{fb} and the timing at which the direction of the current I_s changes (i.e., voltage V_{c0}). FIG. 8 is a chart for explaining details of the drive signals V_{dr1} and V_{dr2} in the normal mode M1.

(102) At time t_0 , for example, the digital control circuit **150** changes the drive signal V_{dr1} to low, and thus the high-side NMOS transistor **31** in FIG. 1 is turned off. Thereafter, at time t_1 at which a time period T_{d1} corresponding to the dead time has elapsed, the digital control circuit **150** changes the drive signal V_{dr2} to high. As a result, the low-side NMOS transistor **32** is turned on.

(103) After the NMOS transistor **32** is turned on, the current I_s gradually decreases and reaches zero at time t_2 . As a result, the comparator **122** in FIG. 2 outputs a low voltage V_{c0} indicating that the current I_s is flowing in the negative direction.

(104) Then, the digital control circuit **150** calculates a time period T_{ah} corresponding to the product of a time period T_{bh} from time t_0 to time t_2 and the feedback voltage V_{fb} , as expressed by Equation (2) as follows:

$$T_{ah}=k \times V_{fb} \times T_{bh} \quad (2),$$

where “k” is a predetermined constant.

(105) At time **t3** at which the time period **Tah** has elapsed since time **t2**, the digital control circuit **150** changes the drive signal **Vdr2** to low, to turn off the NMOS transistor **32**.

(106) At time **t4** at which a time period **Td2** corresponding to the dead time has elapsed since time **t3**, the digital control circuit **150** changes the drive signal **Vdr1** to high. As a result, the high-side NMOS transistor **31** in FIG. **1** is turned on.

(107) After the NMOS transistor **31** is turned on, the current **Is** gradually increases, to reach zero at time **t5**. As a result, the comparator **122** in FIG. **2** outputs a high voltage **Vc0** indicating that the current **Is** is flowing in the positive direction.

(108) Then, the digital control circuit **150** calculates a time period **Tal** corresponding to the product of a time period **Tbl** from time **t3** to time **t5** and the feedback voltage **Vfb**, as expressed by Equation (3) as follows:

$$T_{al}=k \times V_{fb} \times T_{bl} \quad (3),$$

where “k” is the foregoing predetermined constant.

(109) At time **t6** at which the time period **Tal** has elapsed since time **t5**, the digital control circuit **150** changes the drive signal **Vdr1** to low. Further, from time **t6**, the operation from time **t0** to time **t6** is repeated. As such, the switching cycle **Tsw** according to an embodiment of the present disclosure is determined based on the time period **Tah** and the time period **Tal**. Accordingly, the switching cycle **Tsw** increases as the output voltage **Vout** drops and the feedback voltage **Vfb** rises.

(110) Such a relationship as illustrated in FIG. **9** is established between a gain ($=V_{out}/V_{in}$) of the LLC current resonant DC-DC converter **10** and the switching frequency of the NMOS transistors **31** and **32**, for example. In an embodiment of the present disclosure, the switching frequency of the NMOS transistors **31** and **32** are designed to be higher than the resonant frequency of the resonant circuit including the primary coil **L1** and the capacitor **34** in FIG. **1**. In other words, in an embodiment of the present disclosure, the switching frequency of the NMOS transistors **31** and **32** is designed so as to change in a “use area” illustrated in FIG. **9**.

(111) If the output voltage **Vout** of the DC-DC converter **10** drops from the target level, the feedback voltage **Vfb** rises, and thus the switching cycle **Tsw** increases. In this case, the switching frequency of the NMOS transistors **31** and **32** decreases and the gain ($=V_{out}/V_{in}$) increases, resulting in a rise in the output voltage **Vout**.

(112) Meanwhile, if the level of the output voltage **Vout** rises from the target level, the feedback voltage **Vfb** drops, and thus the switching cycle **Tsw** decreases. In this case, the switching frequency of the NMOS transistors **31** and **32** increases and the gain ($=V_{out}/V_{in}$) decreases, resulting in a drop in the output voltage **Vout**. Accordingly, the DC-DC converter **10** outputs the drive signals **Vdr1** and **Vdr2** according to the normal mode **M1**, thereby being able to generate the output voltage **Vout** at the target level.

(113) <<<When Output Voltage **Vout** is Higher than Voltage **V1**>>>

(114) As illustrated in FIG. **6**, when the power consumption of the load **11** decreases and the output voltage **Vout** exceeds the voltage **V1**, for example, the digital control circuit **150** outputs drive signals **Vdr1** and **Vdr2** according to the burst mode **M2**. In an embodiment of the present disclosure, in response to the level of the output voltage **Vout** reaching the voltage **V1**, the feedback voltage **Vfb** reaches the voltage **V11**.

(115) The “burst mode **M2**” according to an embodiment of the present disclosure is an operation mode in which the switching period **T1** during which the NMOS transistors **31** and **32** are continuously switched and the stop period **T2** during which switching of the NMOS transistors **31** and **32** is stopped are repeated in a predetermined cycle **Tx**.

(116) FIG. **10** is a chart illustrating an example of the drive signals **Vdr1** and **Vdr2** according to the burst mode **M2**. The digital control circuit **150** according to an embodiment of the present disclosure outputs the same drive signals **Vdr1** and **Vdr2** as in the normal mode **M1** during the

switching period **T1** in the burst mode **M2**. To be more specific, as illustrated in FIGS. 5 and 8, the digital control circuit **150** acquires the feedback voltage V_{fb} and the timing at which the direction of the current I_s changes (i.e., voltage V_{c0}). Then, the digital control circuit **150** outputs drive signals V_{dr1} and V_{dr2} having the switching cycle T_{sw} obtained using Equations (2) and (3).

(117) Accordingly, in the DC-DC converter **10**, even when the power consumption of the load **11** decreases and the operation mode of the digital control circuit **150** changes from the normal mode **M1** to the burst mode **M2**, the switching method of the NMOS transistors **31** and **32** does not change. As a result, in an embodiment of the present disclosure, even when the operation mode of the digital control circuit **150** changes, an unstable operation of the DC-DC converter **10** can be prevented.

(118) Here, the “switching method” is determined by the switching pattern of the NMOS transistors **31** and **32** and the switching cycle T_{sw} , for example. In an embodiment of the present disclosure, in each of the normal mode **M1** and the burst mode **M2**, the NMOS transistors **31** and **32** have a switching pattern in which the NMOS transistors **31** and **32** are complementarily on and off if the dead time is ignored.

(119) Furthermore, in each of the normal mode **M1** and the burst mode **M2**, the switching cycle T_{sw} is determined based on the same Equations (2) and (3), for example. Accordingly, in an embodiment of the present disclosure, the switching method in the normal mode **M1** and the switching method in the burst mode **M2** are the same. The switching cycle T_{sw} in the normal mode **M1** and the switching cycle T_{sw} in the burst mode **M2** each correspond to a “first cycle”, and the switching cycle T_{sw} in the burst mode **M3** corresponds to a “second cycle”.

(120) FIGS. 11A to 11C are charts for explaining waveforms in the burst mode **M2** when the output voltage V_{out} changes. FIG. 11A is an example of a waveform in the burst mode **M2** when the output voltage V_{out} is the voltage V_1 (i.e., the feedback voltage V_{fb} is the voltage V_{11}). In this case, the digital control circuit **150** sets the switching period **T1** to a time period T_i (e.g., 6 ms), which is an initial value, in a predetermined cycle T_x (e.g., 10 ms), and sets the stop period **T2** to a time period (e.g., 4 ms) obtained by subtracting the time period T_i from the predetermined cycle T_x . That is, in an embodiment of the present disclosure, the stop period **T2** results in $T_x - T_1$ ($=T_x - T_i$).

(121) For example, when the power consumption of the load **11** decreases and the output voltage V_{out} rises from the voltage V_1 (i.e., when the feedback voltage V_{fb} drops from the voltage V_{11}), the digital control circuit **150** gradually reduces the switching period **T1** from the time period T_i (see FIG. 11B).

(122) In an embodiment of the present disclosure, in the burst mode **M2**, the cycle determined by the switching period **T1** and the stop period **T2** is the predetermined cycle T_x , and thus is constant. Accordingly, in response to the switching period **T1** decreases, the power transmitted from the primary side to the secondary side in the DC-DC converter **10** also decreases. As a result, a rise in the output voltage V_{out} can be suppressed.

(123) If the power consumption of the load **11** further decreases and the output voltage V_{out} reaches the voltage V_2 (i.e., if the feedback voltage V_{fb} drops to the voltage V_{12}), the digital control circuit **150** stops gradually reducing the switching period **T1** as illustrated in FIGS. 6 and 11C, for example. To be more specific, when the output voltage V_{out} is in a range of from the voltage V_2 to the voltage V_3 (i.e., when the feedback voltage V_{fb} is in a range of from the voltage V_{12} to the voltage V_{13}), the digital control circuit **150** maintains a time period T_f , which is the switching period **T1** when the feedback voltage V_{fb} is the voltage V_{12} , as the switching period **T1**. Note that the time period T_f , which is the switching period **T1** in the predetermined cycle T_x (e.g., 10 ms), is 2 ms, for example, and thus the stop period **T2** is 8 ms.

(124) As such, in an embodiment of the present disclosure, in response to the output voltage V_{out} exceeding the voltage V_2 , the switching period **T1** is fixed to the time period T_f in the burst mode **M2**. If the switching period **T1** reaches zero and the switching of the NMOS transistors **31** and **32**

is stopped for a long period of time, each coil of the transformer **33** and a parasitic capacitance and the like of the NMOS transistor **31**, **32**, for example, may be discharged. In such a case, when the DC-DC converter **10** restarts the switching operation, unwanted noise may be generated and the operation of the DC-DC converter may become unstable.

(125) In the DC-DC converter **10** according to an embodiment of the present disclosure, even when the output voltage V_{out} exceeds the voltage V_2 , the switching period T_1 does not reach zero, and thus, the operation of the DC-DC converter **10** can be stabilized while noise generation is suppressed.

(126) The switching period T_1 corresponds to a “first time period”, the stop period T_2 corresponds to a “second time period”, and the time period T_f corresponds to a “predetermined time period”. Note that the time period T_f is the switching period T_1 when the level of the output voltage V_{out} reaches the level of the voltage V_2 .

(127) <<<When Output Voltage V_{out} is Higher than Voltage V_3 >>>

(128) In response to the power consumption of the load **11** decreasing and the output voltage V_{out} exceeding the voltage V_3 (i.e., in response to the feedback voltage V_{fb} dropping to the voltage V_{13}), the digital control circuit **150** outputs low drive signals V_{dr1} and V_{dr2} to stop switching the NMOS transistors **31** and **32**, as illustrated in FIG. 6, for example.

(129) ===When Load **11** is in Light Load Condition===

(130) Next, a description is given of a case where the load **11** is in the light load condition, in other words, the power consumption of the load **11** is equal to or less than 3% in FIG. 6. The digital control circuit **150** determines that the power consumption of the load **11** is equal to or less than 3%, based on the voltage V_{ca} .

(131) First, when a predetermined condition is satisfied in the state where the load **11** is in the light load condition, the digital control circuit **150** outputs low drive signals V_{dr1} and V_{dr2} so as to stop switching the NMOS transistors **31** and **32**.

(132) Here, “when the predetermined condition is satisfied” means “when the output voltage V_{out} is higher than the voltage V_3 (i.e., when the feedback voltage V_{fb} is lower than the voltage V_{13})” and “when the output voltage V_{out} is lower than the voltage V_0 (i.e., when the feedback voltage V_{fb} is higher than the voltage V_{10})”.

(133) When the output voltage V_{out} is higher than the voltage V_0 and lower than the voltage V_4 (i.e., when the feedback voltage V_{fb} is lower than the voltage V_{10} and higher than the voltage V_{14}) in a state where the load **11** is in the light load condition, the digital control circuit **150** outputs drive signals V_{dr1} and V_{dr2} according to the foregoing normal mode **M1**.

(134) The level of the voltage V_4 in an embodiment of the present disclosure is higher than the target level of the output voltage V_{out} and lower than the level of the voltage V_1 at which the burst mode **M2** is started. As a result, even when the load **11** is in the light load condition, the DC-DC converter **10** operates in the normal mode **M1** unless the output voltage V_{out} reaches a relatively high voltage V_4 ($<V_1$).

(135) Meanwhile, when the output voltage V_{out} is higher than the voltage V_4 and lower than the voltage V_3 (i.e., when the feedback voltage V_{fb} is lower than the voltage V_{14} and higher than the voltage V_{13}) in the state where the load **11** is in the light load condition, the digital control circuit **150** outputs drive signals V_{dr1} and V_{dr2} according to the burst mode **M3**. Here, the burst mode **M3** is an operation mode in which the switching period T_1 during which the NMOS transistors **31** and **32** are continuously switched and the stop period T_2 during which the switching of the NMOS transistors **31** and **32** is stopped are repeated.

(136) In the burst mode **M3**, unlike the burst mode **M2**, the switching cycle T_{sw} is determined based on the voltage V_w and the input voltage V_{in} , as illustrated in FIG. 5. To be more specific, the digital control circuit **150** in FIG. 2 outputs drive signals V_{dr1} and V_{dr2} , based on the result of detection (voltages V_{c1} and V_{c2}) of the detection circuit **140** illustrated in FIG. 3, in the burst mode **M3**. As a result, as described with reference to FIG. 4, the digital control circuit **150** outputs

a high drive signal Vdr1, in response to the voltage Vc1 going high, and outputs a high drive signal Vdr2, in response to the voltage Vc2 going high.

(137) Further, as illustrated in FIG. 5, the digital control circuit 150 also determines the switching period T1 based on the feedback voltage Vfb when operating in the burst mode M3. FIG. 12 is a chart for explaining the switching period T1 and the stop period T2 when the digital control circuit 150 operates in the burst mode M3.

(138) For example, at time t10 at which the feedback voltage Vfb rises to a voltage V20 at a predetermined level, the digital control circuit 150 generates drive signals Vdr1 and Vdr2 according to the burst mode M3. As a result, the NMOS transistors 31 and 32 are switched.

(139) In response to the NMOS transistors 31 and 32 being driven at time t10, the output voltage Vout rises, and thus the feedback voltage Vfb drops slightly after time t10. Then, at time t11 at which the feedback voltage Vfb drops to a voltage V21 at a predetermined level, for example, the generation of the drive signals Vdr1 and Vdr2 is stopped. Note that a time period from time t10 to time t11 is the “switching period T1”.

(140) As a result, the switching of the NMOS transistors 31 and 32 is also stopped, and thus the output voltage Vout drops. When the feedback voltage Vfb rises slightly after time t11 and reaches the voltage V20 at time t12, for example, the drive signals Vdr1 and Vdr2 according to the burst mode M3 are generated. As a result, the NMOS transistors 31 and 32 are driven again at this timing. Note that a time period from time t11 to time t12 is the “stop period T2”.

(141) From time t12, the operation from time t10 to time t12 is repeated. As such, the digital control circuit 150 operates in the burst mode M3, during the switching period T1 according to the feedback voltage Vfb. Accordingly, in an embodiment of the present disclosure, it is possible to suppress a rise in the output voltage Vout in the state where the load 11 is in the light load condition, while suppressing the power consumption of the DC-DC converter 10. Note that the voltage V20 is a predetermined voltage that is close to the voltage V14 and lower than the voltage V14. The voltage V21 is a predetermined voltage that is close to the voltage V13 and higher than the voltage V14.

Example of Operation Mode when Power Consumption of Load 11 Decreases

(142) FIG. 13 is a chart illustrating an example of the operation mode of the digital control circuit 150 when the power consumption of the load 11 decreases. Here, it is assumed that before time t30, the DC-DC converter 10 operates in the normal mode M1 and generates an output voltage Vout at the target level.

(143) At time t30, upon reduction in the power consumption of the load 11, the output voltage Vout rises above the target level. Here, it is assumed that the load 11 is under a load condition heavier than in the light load condition (i.e., 3% of the rating). Upon rise in the output voltage Vout, the feedback voltage Vfb drops from the voltage V15 corresponding to the target level.

(144) Then, in response to the feedback voltage Vfb reaching the voltage V11 at time t31, the operation mode of the digital control circuit 150 changes from the normal mode M1 to the burst mode M2. As a result, as illustrated in FIG. 11A, the switching period T1 and the stop period T2 are repeated in the predetermined cycle Tx.

(145) Since the feedback voltage Vfb gradually drops from time t31, the digital control circuit 150 gradually reduces the switching period T1 in the predetermined cycle Tx, as illustrated in FIG. 11B. As a result, the power transmitted from the primary side to the secondary side in the DC-DC converter 10 is reduced, and thus a rise in the output voltage Vout is suppressed, for example.

(146) In response to the feedback voltage Vfb reaching the voltage V12 at time t32, the digital control circuit 150 sets the switching period T1 to a minimum time period Tf in the predetermined cycle Tx, as illustrated in FIG. 11C. As a result, even when the feedback voltage Vfb drops from the voltage V12 from time t32, the switching period T1 does not decrease.

(147) Thereafter, upon drop in the output voltage Vout, the feedback voltage Vfb rises and reaches the voltage V12 at time t33. As a result, from time t33, the digital control circuit 150 increases the

switching period **T1** according to a rise in the feedback voltage **Vfb**. Then, in response to the feedback voltage **Vfb** reaching the voltage **V11** at time **t34**, the operation mode of the digital control circuit **150** changes from the burst mode **M2** to the normal mode **M1**.

(148) As described above, when the output voltage **Vout** is higher than the target level in the normal mode **M1**, the digital control circuit **150** outputs the drive signals **Vdr1** and **Vdr2** for causing the output voltage **Vout** to reach the target level. As a result, the output voltage **Vout** of the DC-DC converter **10** reaches the target level.

SUMMARY

(149) The DC-DC converter **10** according to an embodiment of the present disclosure has been described above. The digital control circuit **150** of the control IC **50** outputs the drive signals **Vdr1** and **Vdr2** according to the normal mode **M1** when the output voltage **Vout** is lower than the voltage **V1**, and outputs the drive signals **Vdr1** and **Vdr2** according to the burst mode **M2** when the output voltage **Vout** is higher than the voltage **V1** (see FIG. 6). Then, as illustrated in FIG. 5, for example, in each of the normal mode **M1** and the burst mode **M2**, the switching cycle **Tsw** of the drive signals **Vdr1** and **Vdr2** is generated by the same method. Then, as the output voltage **Vout** rises above the voltage **V1**, the digital control circuit **150** reduces the switching period **T1** as illustrated in FIG. 11B, for example. Accordingly, in an embodiment of the present disclosure, even when the condition of the load **11** changes and the operation mode of the DC-DC converter **10** changes, the DC-DC converter **10** can be stably operated.

(150) Further, in response to the output voltage **Vout** reaching the voltage **V2** higher than the voltage **V1**, the digital control circuit **150** stops reducing the switching period **T1** as illustrated in FIG. 11C, for example. Accordingly, even when the output voltage **Vout** exceeding the voltage **V2**, the digital control circuit **150** fixes the switching period **T1** in the burst mode **M2** to the time period **Tf**. As a result, in an embodiment of the present disclosure, it is possible to prevent switching of the NMOS transistors **31** and **32** from being stopped while suppressing a rise in the output voltage **Vout**.

(151) The digital control circuit **150** outputs low drive signals **Vdr1** and **Vdr2** to stop switching the NMOS transistors **31** and **32**, in response to the output voltage **Vout** reaching the voltage **V3**. As a result, the output voltage **Vout** can be prevented from rising abnormally.

(152) The digital control circuit **150** may output the drive signals **Vdr1** and **Vdr2** according to the burst mode **M2** different from the burst mode **M1**, in response to the load **11** entering the light load condition, for example. Such a case uses burst modes that are different depending on whether the load **11** is in the light load condition or not, and thus the DC-DC converter **10** can select the optimum burst mode according to the condition of the load **11**.

(153) The digital control circuit **150** operates in the burst mode **M2**, in response to the load **11** entering the light load condition and the output voltage **Vout** exceeding the voltage **V4**.

Accordingly, even in a state where the load **11** is in the light load condition, the digital control circuit **150** operates in the normal mode **M1** when the output voltage **Vout** is low. This can prevent unnecessary switching of the operation mode.

(154) The digital control circuit **150** operates in the burst mode **M2**, in response to the power consumption of the load **11** being equal to or less than 5% (here, e.g., 3%) of the rated power. In an embodiment of the present disclosure, it is possible to set conditions for using the burst mode **M2** different from the burst mode **M1**, particularly in the state where the load **11** is in the light load condition. Thus, it is possible to particularly improve the conversion efficiency of the DC-DC converter **10** under the light load condition.

(155) The digital control circuit **150** determines the switching cycle **Tsw**, based on the voltage **Vw** at the terminal **VW**, when operating in the burst mode **M2**.

(156) The digital control circuit **150** determines the switching cycle **Tsw**, based on the voltage **Vw** and the input voltage **Vin**, when operating in the burst mode **M2**. For example, when the input voltage **Vin** of the DC-DC converter **10** rises, the voltages generated at the primary coil and the

auxiliary coil L4 also rise, and the power transmitted to the secondary side also increases.

Accordingly, in an embodiment of the present disclosure, power can be appropriately transmitted to the secondary side in accordance with the level of the input voltage V_{in} , in the burst mode M2.

(157) The present disclosure is directed to provision of a switching control circuit capable of stably operating a power supply circuit when a load condition changes.

(158) According to the present disclosure, it is possible to provide a switching control circuit capable of stably operating a power supply circuit when a load condition changes.

(159) Embodiment(s) of the present disclosure described above is/are simply to facilitate understanding of the present disclosure and is/are not in any way to be construed as limiting the present disclosure. The present disclosure may variously be changed or altered without departing from its essential features and encompass equivalents thereof.

Claims

1. A switching control circuit for a power supply circuit that includes a transformer including a primary coil and a secondary coil, and first and second transistors configured to control a current flowing through the primary coil, the power supply circuit being configured to generate an output voltage at a target level, the switching control circuit configured to control switching of the first and second transistors, the switching control circuit comprising: a drive signal output circuit configured to output drive signals according to a normal mode, when the output voltage is lower than a first level, and output the drive signals according to a first burst mode, when the output voltage is higher than the first level; and a driver circuit configured to switch the first and second transistors in response to the drive signals, wherein the normal mode is a mode in which the first and second transistors are continuously switched, in a first cycle based on a feedback voltage corresponding to the output voltage, the first burst mode is a mode in which a first time period, during which the first and second transistors are continuously switched in the first cycle, and a second time period, during which the switching of the first and second transistors is stopped, are repeated in a predetermined cycle, and the drive signal output circuit reduces the first time period in the predetermined cycle, in response to the output voltage rising above the first level.
2. The switching control circuit according to claim 1, wherein the drive signal output circuit sets the first time period to a predetermined time period, in response to the output voltage reaching a second level higher than the first level.
3. The switching control circuit according to claim 2, wherein the drive signal output circuit outputs the drive signals to stop the switching of the first and second transistors, in response to the output voltage reaching a third level higher than the second level.
4. The switching control circuit according to claim 1, wherein the power supply circuit further includes a first capacitor coupled in series with the primary coil, and drives a load; the switching control circuit further includes a load detection circuit configured to detect a condition of the load of the power supply circuit, based on a resonant current of a resonant circuit that is formed by the primary coil and the first capacitor; and the drive signal output circuit outputs the drive signals according to a second burst mode different from the first burst mode, in response to power consumed by the load becoming smaller than a first value.
5. The switching control circuit according to claim 4, wherein the drive signal output circuit outputs the drive signals according to the second burst mode, in response to the power consumed by the load becoming smaller than the first value and the output voltage exceeding another level lower than the first level.
6. The switching control circuit according to claim 4, wherein the load has a rated power; and the first value is equal to or less than 5% of the rated power of the load.
7. The switching control circuit according to claim 4, wherein the transformer includes an auxiliary coil, the switching control circuit is an integrated circuit having a first terminal configured to

receive a voltage corresponding to a voltage at the auxiliary coil, and the drive signals outputted by the drive signal output circuit according to the second burst mode have a second cycle based on the voltage at the first terminal.

8. The switching control circuit according to claim 7, further comprising: a second terminal configured to receive an input voltage of the power supply circuit, wherein the second cycle of the outputted drive signals is based on both the voltage at the first terminal and a voltage at the second terminal.

9. A power supply circuit configured to generate an output voltage at a target level, the power supply circuit comprising: a transformer including a primary coil and a secondary coil; first and second transistors configured to control a current flowing through the primary coil; and a switching control circuit configured to control switching of the first and second transistors, wherein the switching control circuit includes: a drive signal output circuit configured to output drive signals according to a normal mode, when the output voltage is lower than a first level, and output the drive signals according to a first burst mode, when the output voltage is higher than the first level, and a driver circuit configured to switch the first and second transistors, in response to the drive signals, wherein the normal mode is a mode in which the first and second transistors are continuously switched, in a first cycle based on a feedback voltage corresponding to the output voltage, and the first burst mode is a mode in which a first time period, during which the first and second transistors are continuously switched in the first cycle, and a second time period, during which the switching of the first and second transistors is stopped, are repeated in a predetermined cycle; and the drive signal output circuit reduces the first time period in the predetermined cycle, in response to the output voltage rising above the first level.
